

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458729.8231 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.213 +/- 0.06
Transit depth (Rp/Rs)^2	4.52 +/- 2.57 [%]
Orbital Inclination (inc)	82.6 +/- 2.55
Airmass coefficient 1 (a1)	0.789 +/- 0.017
Airmass coefficient 2 (a2)	0.208 +/- 0.079
Scatter in the residuals of the lightcurve fit is	2.13 %
Best Comparison Star	#3 - [298, 452]
Optimal Aperture	44.75
Optimal Annulus	139.13
Transit Duration (day)	0.018 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.213 $\pm$ 0.06 vs 0.1646 (deviation 0.81 σ)
Duration fit vs published	0.018 d vs 0.0758 d (deviation 4.45 σ)
Mid-time O-C	12.5 min
Depth SNR	3.6
Residual scatter	2.13 %

Light curve

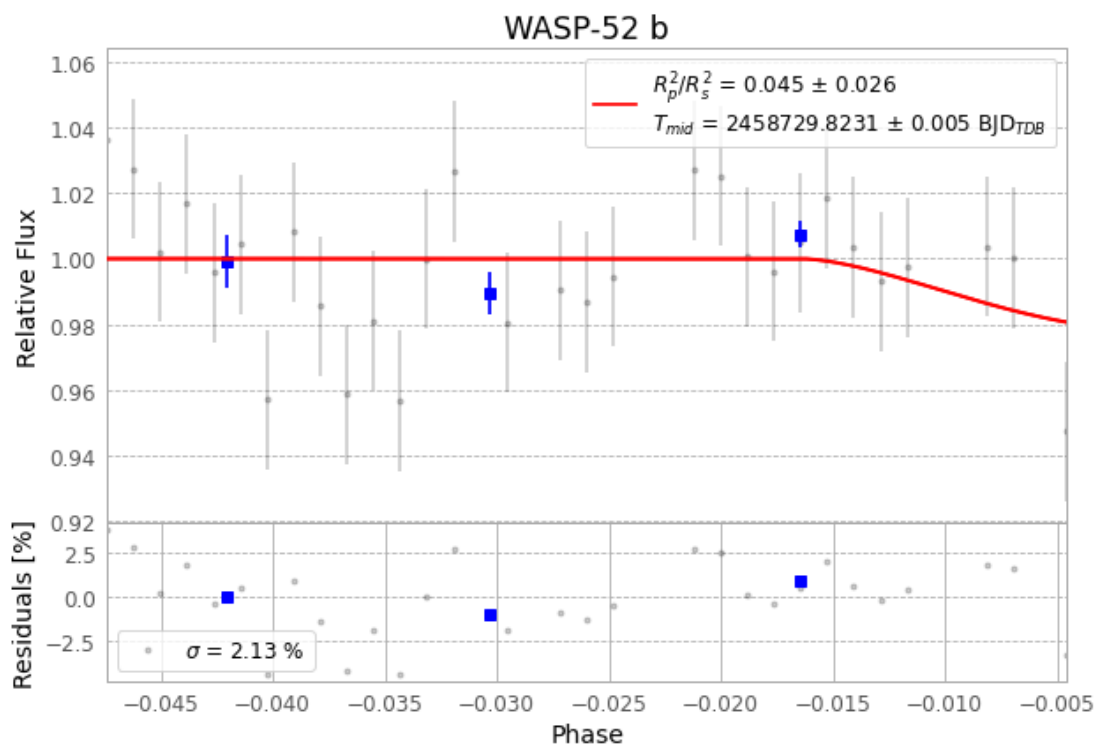


Figure 1 — FinalLightCurve_WASP-52 b_2019-09-02.png (SHA-256 e535840b43cb3c65...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 449], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3a995069d1a2962d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52190903053612.FITS → WASP-52190903092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-190903.log (74cd4894c32f...), FinalLightCurve_WASP-52 b_2019-09-02.png (e535840b43cb...), FinalLightCurve_WASP-52 b_2019-09-02.csv (47a0b0b633aa...)

Compute resources

Wall time 455.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:56Z.